

StomaML: Computer Vision for Stomatal Density Detection in Birch Leaves

Valtteri Ahola¹, Soumyadipta Kar¹, Mia Linblad¹, Miika Mäki¹, and Markus Tarma¹

¹*University of Helsinki*

March 3, 2026 3:43pm

Introduction

Stomatal sizes, shapes and densities fluctuate by environmental factors [1, 2, 3]. Analysing stomatal characteristics is crucial for understanding plant responses to environmental and climate changes [1, 4, 5].

Several attempts have been made to automatate stomatal detection with computer vision since the 1980s [6, 7, 8, 9]. Recently, deep learning approaches have improved the accuracy and made the models more general [10, 11, 12]. Still, the tools tend to be species specific, do not maintain acceptable accuracy for species they have not been trained with or contain only limited morphological traits [13].

Images of birch leaves cannot currently be handled automatically with existing solutions. Instead, the stomata for birch tree leaves must be counted manually which is extremely time consuming and forms a significant bottleneck in research workflows [11, 13]. This is because the topography of birch leaf surfaces are uneven, making parts of the images blurry even if one would try to stack multiple images. In addition, some leaf structures, like veins or trichomes cannot contain stomata, and should be excluded from a valid countable area [14]. In order to be able to identify stomata and achieve a meaningful stomatal density, a targeted solution for birch leaves is needed.

We have developed a standalone web application, StomaML, that automatically detects and quantifies stomata in images of birch leaves. Users can upload batches of leaf images, and the system outputs a CSV file containing stomata counts per image, countable area of stomata (excluding veins and artifacts), stomatal density & stomata size measurements, as well as comprehensive metadata about each object identified per image. With automation the analysis of tens of images could be done quickly, and a subsequent correlation study would be made significantly easier.

Data

We used 376 microscope images that were magnified 20-fold. The images had been taken and were provided by the client, Department of Organismal and Evolutionary Biology at Faculty of Biological and Environmental Sciences at the University of Helsinki.

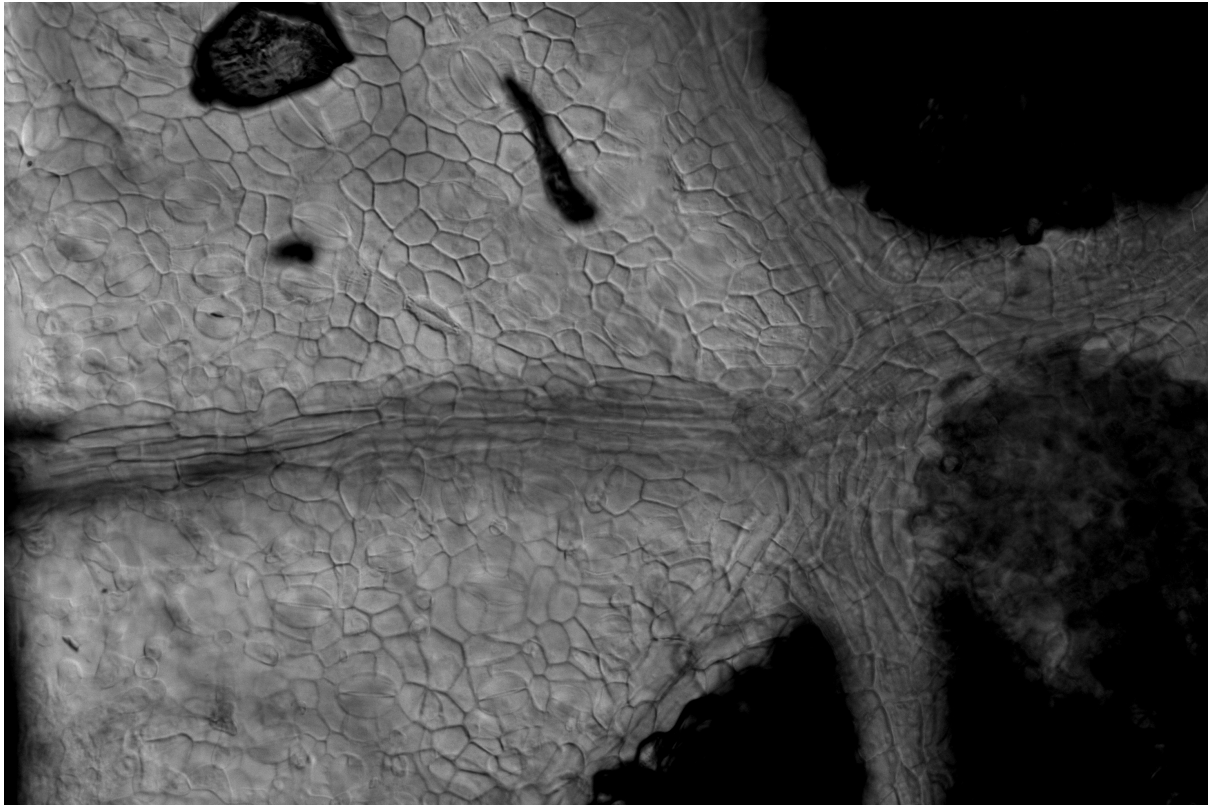


FIGURE 1. EXAMPLE IMAGE DEMONSTRATING STOMATA, TRICHOMES, VEINS, 'NOTHING' AND BLURRY AREAS

Each image had an average of 40 stomatal pores. In addition, there are trichomes, veins, blurred areas as well as what we call *nothing* —or areas outside of the leaves or small holes. In total, we identified XXXXX objects during training. Please refer to Figure 1 for an example image.

Methods

We used Roboflow for annotating and training our computer vision models [15]. Roboflow makes the tedious annotation somewhat more tolerable: There is a possibility to add and manage multiple object classes, a smart selection tool that runs on SAM 3 [16], and convenient way of assigning images for annotation among project members. After annotating the first 100 images, we used the existing models to pre-annotate the rest of the pictures. Still the annotation took a large share of our total time with the project. Luckily, the client could mobilize five people to help us annotate the non-countable area and do sanity checks for the stomatal detection.

Roboflow also makes it convenient to train and test the models. The model could be exported to deployment directly from roboflow.

Results

Discussion

Our group created a web application that automatically counts stomatal counts, densities, sizes, lengths and non-countable area.

References

- [1] A. M. Hetherington and F. I. Woodward, “The role of stomata in sensing and driving environmental change,” *Nature*, vol. 424, no. 6951, pp. 901–908, Aug. 2003, ISSN: 1476-4687. DOI: 10.1038/nature01843. Accessed: Mar. 3, 2026. [Online]. Available: <https://www.nature.com/articles/nature01843>.
- [2] S. Casson and J. E. Gray, “Influence of environmental factors on stomatal development,” *New Phytologist*, vol. 178, no. 1, pp. 9–23, 2008, ISSN: 1469-8137. DOI: 10.1111/j.1469-8137.2007.02351.x. Accessed: Mar. 3, 2026. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1469-8137.2007.02351.x>.
- [3] L. T. Bertolino, R. S. Caine, and J. E. Gray, “Impact of Stomatal Density and Morphology on Water-Use Efficiency in a Changing World,” *Frontiers in Plant Science*, vol. 10, Mar. 6, 2019, ISSN: 1664-462X. DOI: 10.3389/fpls.2019.00225. Accessed: Mar. 3, 2026. [Online]. Available: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2019.00225/full>.
- [4] W. Yan, Y. Zhong, and Z. Shangguan, “Contrasting responses of leaf stomatal characteristics to climate change: A considerable challenge to predict carbon and water cycles,” *Global Change Biology*, vol. 23, no. 9, pp. 3781–3793, 2017, ISSN: 1365-2486. DOI: 10.1111/gcb.13654. Accessed: Mar. 3, 2026. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/gcb.13654>.
- [5] X. Liang et al., “Stomatal responses of terrestrial plants to global change,” *Nature Communications*, vol. 14, no. 1, p. 2188, Apr. 17, 2023, ISSN: 2041-1723. DOI: 10.1038/s41467-023-37934-7. Accessed: Mar. 3, 2026. [Online]. Available: <https://www.nature.com/articles/s41467-023-37934-7>.
- [6] K. Omasa and M. Onoe, “Measurement of Stomatal Aperture by Digital Image Processing,” *Plant and Cell Physiology*, vol. 25, no. 8, pp. 1379–1388, Dec. 1, 1984, ISSN: 0032-0781. DOI: 10.1093/oxfordjournals.pcp.a076848. Accessed: Mar. 3, 2026. [Online]. Available: <https://doi.org/10.1093/oxfordjournals.pcp.a076848>.
- [7] G. Karabourniotis, D. Tzobanoglou, D. Nikolopoulos, and G. Liakopoulos, “Epicuticular Phenolics Over Guard Cells: Exploitation for in situ Stomatal Counting by Fluorescence Microscopy and Combined Image Analysis,” *Annals of Botany*, vol. 87, no. 5, pp. 631–639, May 1, 2001, ISSN: 0305-7364. DOI: 10.1006/anbo.2001.1386. Accessed: Mar. 3, 2026. [Online]. Available: <https://doi.org/10.1006/anbo.2001.1386>.
- [8] H. Laga, F. Shahinnia, and D. Fleury, “Image-based plant stomata phenotyping,” in *2014 13th International Conference on Control Automation Robotics & Vision (ICARCV)*, Dec. 2014, pp. 217–222. DOI: 10.1109/ICARCV.2014.7064307. Accessed: Mar. 3, 2026. [Online]. Available: <https://ieeexplore.ieee.org/document/7064307>.

- [9] K. T. N. Duarte, M. A. G. D. Carvalho, and P. S. Martins, “Segmenting High-quality Digital Images of Stomata using the Wavelet Spot Detection and the Watershed Transform:” in *Proceedings of the 12th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications*, Porto, Portugal: SCITEPRESS - Science and Technology Publications, 2017, pp. 540–547, ISBN: 978-989-758-225-7 978-989-758-226-4 978-989-758-227-1. doi: 10.5220/0006168105400547. Accessed: Mar. 3, 2026. [Online]. Available: <http://www.scitepress.org/DigitalLibrary/Link.aspx?doi=10.5220/0006168105400547>.
- [10] K. C. Fetter, S. Eberhardt, R. S. Barclay, S. Wing, and S. R. Keller, “StomataCounter: A neural network for automatic stomata identification and counting,” *New Phytologist*, vol. 223, no. 3, pp. 1671–1681, 2019, ISSN: 1469-8137. doi: 10.1111/nph.15892. Accessed: Mar. 3, 2026. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/nph.15892>.
- [11] N. Sai et al., “StomaAI: An efficient and user-friendly tool for measurement of stomatal pores and density using deep computer vision,” *The New Phytologist*, vol. 238, no. 2, pp. 904–915, Apr. 2023, ISSN: 1469-8137. doi: 10.1111/nph.18765. PMID: 36683442.
- [12] J. A. Gibbs, L. Mcausland, C. A. Robles-Zazueta, E. H. Murchie, and A. J. Burgess, “A Deep Learning Method for Fully Automatic Stomatal Morphometry and Maximal Conductance Estimation,” *Frontiers in Plant Science*, vol. 12, Dec. 2, 2021, ISSN: 1664-462X. doi: 10.3389/fpls.2021.780180. Accessed: Mar. 3, 2026. [Online]. Available: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2021.780180/full>.
- [13] J. A. Gibbs and A. J. Burgess, “Application of deep learning for the analysis of stomata: A review of current methods and future directions,” *Journal of Experimental Botany*, vol. 75, no. 21, pp. 6704–6718, Nov. 15, 2024, ISSN: 0022-0957. doi: 10.1093/jxb/erae207. Accessed: Mar. 3, 2026. [Online]. Available: <https://doi.org/10.1093/jxb/erae207>.
- [14] E. Valkama, J.-P. Salminen, J. Koricheva, and K. Pihlaja, “Comparative Analysis of Leaf Trichome Structure and Composition of Epicuticular Flavonoids in Finnish Birch Species,” *Annals of Botany*, vol. 91, no. 6, pp. 643–655, May 1, 2003, ISSN: 0305-7364. doi: 10.1093/aob/mcg070. Accessed: Mar. 3, 2026. [Online]. Available: <https://doi.org/10.1093/aob/mcg070>.
- [15] B. Dwyer, J. Nelson, T. Hansen, et al., *Roboflow*, version 1.0, 2024. [Online]. Available: <https://roboflow.com>.
- [16] N. Carion et al. “SAM 3: Segment Anything with Concepts.” arXiv: 2511.16719 [cs], Accessed: Feb. 25, 2026. [Online]. Available: <http://arxiv.org/abs/2511.16719>, pre-published.